

Week 4

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pranav_chowdary@Pranav: ~  
pranav_chowdary@Pranav:~$ nano q4.c  
pranav_chowdary@Pranav:~$ gcc q4.c -o q4  
./q4  
Child 1: PID = 1246  
Child 2: PID = 1247  
Child 3: PID = 1248  
Child 1 finished  
Child 2 finished  
Child 3 finished  
All children completed  
pranav_chowdary@Pranav:~$ nano zombie.c  
pranav_chowdary@Pranav:~$ gcc zombie.c -o zombie  
pranav_chowdary@Pranav:~$ ./zombie  
Parent PID: 1292  
Child PID: 1293  
Child finished  
ps -el | grep zombie  
Zombie removed  
pranav_chowdary@Pranav:~$ ps -el | grep zombie  
pranav_chowdary@Pranav:~$ wait(NULL);  
-bash: syntax error near unexpected token 'NULL'  
pranav_chowdary@Pranav:~$ gcc q1.c -o q1  
gcc q2.c -o q2  
gcc q3.c -o q3  
gcc q4.c -o q4  
gcc zombie.c -o zombie  
pranav_chowdary@Pranav:~$ ./q1  
./q2  
./q3  
./q4  
./zombie  
Enter Linux command: ls  
Parent PID: 1401  
Child PID: 1404  
a.out copy.txt main.c q1 q1.c q2 q2.c q3 q3.c q4 q4.c source.txt zombie zombie.c
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pranav_chowdary@Pranav: ~  
Zombie removed  
pranav_chowdary@Pranav:~$ ps -el | grep zombie  
pranav_chowdary@Pranav:~$ wait(NULL);  
-bash: syntax error near unexpected token 'NULL'  
pranav_chowdary@Pranav:~$ gcc q1.c -o q1  
gcc q2.c -o q2  
gcc q3.c -o q3  
gcc q4.c -o q4  
gcc zombie.c -o zombie  
pranav_chowdary@Pranav:~$ ./q1  
./q2  
./q3  
./q4  
./zombie  
Enter Linux command: ls  
Parent PID: 1401  
Child PID: 1404  
a.out copy.txt main.c q1 q1.c q2 q2.c q3 q3.c q4 q4.c source.txt zombie zombie.c  
Child completed  
File copied successfully  
Parent PID: 1416  
Child PID: 1417  
Child PID: 1417  
Parent PID: 1416  
Child finished  
Parent finished  
Child 1: PID = 1428  
Child 2: PID = 1429  
Child 3: PID = 1430  
Child 1 finished  
Child 2 finished  
Child 3 finished  
All children completed  
Parent PID: 1438  
Child PID: 1439
```

```
pranav_chowdary@Pranav: ~  
GNU nano 8.7.1 q4.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int i;  
  
    for (i = 0; i < 3; i++)  
    {  
        if (fork() == 0)  
        {  
            printf("Child %d: PID = %d\n", i + 1, getpid());  
            sleep(i + 1);  
            printf("Child %d finished\n", i + 1);  
            return 0;  
        }  
    }  
  
    for (i = 0; i < 3; i++)  
    {  
        wait(NULL);  
    }  
  
    printf("All children completed\n");  
    return 0;  
}
```

[Read 28 lines]

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